

Supplementary Material

Full benchmark tables for FEELIN Phase 1 frozen-probe report

2026-05-27

1 Chance baselines

Table 1: Chance baselines per task, by dummy-predictor strategy. Stratified rows are averaged over 3 seeds.

Task	Head	μ	σ	n
A_binary	most_frequent	0.5000	0.0000	5
A_binary	stratified	0.4795	0.0301	15
A_reg	mean	0.0000	0.0000	5
A_reg	median	0.0000	0.0000	5
V_binary	most_frequent	0.5000	0.0000	5
V_binary	stratified	0.5053	0.0329	15
V_reg	mean	0.0000	0.0000	5
V_reg	median	0.0000	0.0000	5

2 Per-task full benchmark

All cells are reported. Within each category (brain, video) rows are sorted by the main metric (AUROC for binary, Pearson r for regression). The n column shows the number of underlying fold $\times$ seed (or, for per_subject mode, fold $\times$ seed $\times$ subject) fits that were averaged.

2.1 V_binary

Table 2: Full benchmark for V_binary (Valence binary classification (extreme Q4 vs Q1)). Main metric: AUROC. Cells sorted by main metric within category.

#	Feature	Init	Padding	Head	Mode	AUROC		AUPRC		balAcc		n
						μ	σ	μ	σ	μ	σ	
<i>Brain models (fMRI input)</i>												
1	ROI_Schaefer400Tian50	n/a	time_mean	linear	pooled	0.7889	0.0119	0.7940	0.0099	0.7199	0.0107	5
2	ROI_Schaefer400Tian50	n/a	time_mean	linear	per_subject	0.7885	0.0263	0.7979	0.0248	0.7229	0.0272	25
3	ROI_Schaefer400Tian50	n/a	time_mean	mlp	pooled	0.7626	0.0142	0.7569	0.0243	0.6998	0.0237	5
4	ROI_Schaefer400Tian50	n/a	time_mean	mlp	per_subject	0.7601	0.0319	0.7611	0.0415	0.7058	0.0275	25

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#	Feature	Init	Padding	Head	Mode	AUROC		AUPRC		balAcc		n
						μ	σ	μ	σ	μ	σ	
5	Brain-JEPA	resting	zero	linear	per_subject	0.7402	0.0365	0.7571	0.0293	0.6797	0.0350	25
6	Brain-JEPA	resting	zero	linear	pooled	0.7376	0.0194	0.7411	0.0132	0.6768	0.0180	5
7	NeuroSTORM	resting	mean	linear	per_subject	0.7292	0.0361	0.7402	0.0371	0.6657	0.0342	25
8	Brain-JEPA	resting	zero	mlp	pooled	0.7083	0.0178	0.7092	0.0184	0.6576	0.0160	5
9	NeuroSTORM	resting	mean	mlp	per_subject	0.7081	0.0398	0.7174	0.0445	0.6576	0.0343	25
10	NeuroSTORM	resting	mean	linear	pooled	0.7019	0.0190	0.7110	0.0229	0.6454	0.0150	5
11	Brain-JEPA	scratch	zero	linear	pooled	0.6957	0.0178	0.6977	0.0214	0.6476	0.0216	5
12	SwiFT_UAH_202M	scratch	zero	linear	per_subject	0.6911	0.0368	0.7102	0.0333	0.6413	0.0346	25
13	Brain-JEPA	resting	zero	mlp	per_subject	0.6900	0.0320	0.7001	0.0312	0.6262	0.0381	25
14	SwiFT_UAH_202M	resting	zero	linear	per_subject	0.6899	0.0360	0.7079	0.0411	0.6377	0.0386	25
15	NeuroSTORM	resting	mean	mlp	pooled	0.6898	0.0273	0.6936	0.0241	0.6415	0.0221	5
16	SwiFT_NewE96	resting	zero	linear	pooled	0.6884	0.0166	0.7075	0.0141	0.6399	0.0113	5
17	SwiFT_NewE192	resting	zero	linear	pooled	0.6863	0.0221	0.7077	0.0135	0.6342	0.0101	5
18	SwiFT_NewE96	resting	mean	linear	pooled	0.6860	0.0177	0.7010	0.0224	0.6300	0.0103	10
19	SwiFT_NewE96	resting	spatial_only	linear	pooled	0.6848	0.0187	0.7008	0.0238	0.6334	0.0111	5
20	SwiFT_NewE96	resting	cyclic_replicate	linear	pooled	0.6835	0.0173	0.6989	0.0226	0.6315	0.0167	5
21	Brain-JEPA	scratch	zero	linear	per_subject	0.6751	0.0317	0.6778	0.0395	0.6320	0.0292	25
22	SwiFT_NewE96	resting	mean	linear	per_subject	0.6711	0.0343	0.6931	0.0382	0.6233	0.0319	50
23	SwiFT_UAH_5M	scratch	zero	linear	per_subject	0.6700	0.0356	0.6929	0.0371	0.6239	0.0285	25
24	SwiFT_NewE96	resting	spatial_only	linear	per_subject	0.6696	0.0333	0.6909	0.0364	0.6258	0.0280	25
25	SwiFT_NewE96	resting	cyclic_replicate	linear	per_subject	0.6677	0.0334	0.6906	0.0362	0.6177	0.0316	25
26	SwiFT_UAH_51M	scratch	zero	linear	per_subject	0.6660	0.0360	0.6805	0.0386	0.6206	0.0265	25
27	SwiFT_UAH_202M	resting	zero	linear	pooled	0.6654	0.0058	0.6739	0.0085	0.6227	0.0064	5
28	SwiFT_NewE96	resting	zero	linear	per_subject	0.6649	0.0290	0.6852	0.0266	0.6204	0.0305	25
29	SwiFT_UAH_202M	scratch	zero	linear	pooled	0.6631	0.0127	0.6833	0.0189	0.6254	0.0116	5
30	SwiFT_UAH_202M	scratch	zero	mlp	per_subject	0.6630	0.0378	0.6909	0.0370	0.5934	0.0427	25
31	SwiFT_UAH_51M	resting	zero	linear	per_subject	0.6618	0.0379	0.6799	0.0318	0.6150	0.0382	25
32	SwiFT_NewE36	resting	zero	linear	pooled	0.6615	0.0272	0.6799	0.0201	0.6185	0.0229	5
33	SwiFT_NewE36	resting	zero	linear	per_subject	0.6590	0.0396	0.6814	0.0390	0.6199	0.0350	25
34	SwiFT_UAH_51M	resting	zero	linear	pooled	0.6586	0.0156	0.6822	0.0104	0.6204	0.0088	5
35	SwiFT_NewE192	scratch	zero	linear	per_subject	0.6583	0.0333	0.6904	0.0367	0.6138	0.0252	25
36	SwiFT_NewE96	scratch	mean	linear	per_subject	0.6571	0.0385	0.6830	0.0412	0.6101	0.0311	50
37	SwiFT_NewE96	scratch	zero	linear	per_subject	0.6546	0.0426	0.6874	0.0373	0.6109	0.0383	25
38	SwiFT_NewE96	scratch	spatial_only	linear	per_subject	0.6544	0.0409	0.6803	0.0456	0.6110	0.0336	25
39	SwiFT_NewE96	scratch	cyclic_replicate	linear	per_subject	0.6537	0.0374	0.6816	0.0373	0.6084	0.0304	25
40	SwiFT_UAH_202M	scratch	zero	mlp	pooled	0.6531	0.0137	0.6791	0.0060	0.6077	0.0222	5
41	SwiFT_NewE96	scratch	replicate	linear	per_subject	0.6500	0.0300	0.6771	0.0274	0.6062	0.0251	25
42	NeuroSTORM	scratch	mean	linear	per_subject	0.6497	0.0477	0.6797	0.0418	0.6092	0.0397	25
43	SwiFT_NewE192	resting	zero	linear	per_subject	0.6464	0.0367	0.6736	0.0368	0.6115	0.0296	25
44	SwiFT_NewE96	resting	replicate	linear	pooled	0.6462	0.0198	0.6606	0.0191	0.6079	0.0157	5
45	SwiFT_UAH_5M	scratch	zero	mlp	per_subject	0.6421	0.0396	0.6713	0.0419	0.5951	0.0335	25
46	NeuroSTORM	scratch	mean	linear	pooled	0.6410	0.0260	0.6592	0.0238	0.6047	0.0224	5
47	SwiFT_NewE96	scratch	mean	linear	pooled	0.6349	0.0161	0.6599	0.0089	0.5934	0.0185	10
48	SwiFT_NewE96	scratch	cyclic_replicate	linear	pooled	0.6331	0.0083	0.6592	0.0052	0.5918	0.0083	5
49	SwiFT_UAH_51M	scratch	zero	linear	pooled	0.6325	0.0217	0.6511	0.0112	0.5987	0.0273	5
50	SwiFT_NewE96	scratch	mean	mlp	per_subject	0.6315	0.0386	0.6563	0.0372	0.5937	0.0360	50
51	SwiFT_NewE96	scratch	zero	linear	pooled	0.6314	0.0162	0.6593	0.0178	0.5925	0.0158	5
52	SwiFT_NewE96	scratch	spatial_only	linear	pooled	0.6303	0.0184	0.6551	0.0117	0.5874	0.0205	5
53	SwiFT_NewE192	scratch	zero	mlp	per_subject	0.6302	0.0367	0.6633	0.0394	0.5831	0.0367	25
54	SwiFT_UAH_5M	scratch	zero	linear	pooled	0.6258	0.0070	0.6475	0.0110	0.5883	0.0081	5
55	SwiFT_NewE96	scratch	replicate	mlp	per_subject	0.6251	0.0371	0.6446	0.0407	0.5865	0.0233	25
56	SwiFT_NewE96	scratch	cyclic_replicate	mlp	per_subject	0.6250	0.0408	0.6462	0.0414	0.5899	0.0400	25
57	SwiFT_NewE96	scratch	replicate	linear	pooled	0.6250	0.0186	0.6596	0.0183	0.5880	0.0165	5
58	SwiFT_NewE192	scratch	zero	linear	pooled	0.6246	0.0191	0.6554	0.0131	0.5899	0.0221	5
59	SwiFT_NewE96	resting	replicate	linear	per_subject	0.6242	0.0423	0.6490	0.0327	0.5908	0.0358	25
60	SwiFT_NewE96	scratch	cyclic_replicate	mlp	pooled	0.6233	0.0087	0.6403	0.0063	0.5865	0.0092	5
61	SwiFT_UAH_51M	scratch	zero	mlp	per_subject	0.6229	0.0339	0.6484	0.0360	0.5854	0.0255	25
62	NeuroSTORM	scratch	mean	mlp	per_subject	0.6218	0.0550	0.6450	0.0475	0.5836	0.0453	25
63	SwiFT_NewE36	scratch	zero	linear	per_subject	0.6215	0.0433	0.6537	0.0397	0.5895	0.0318	25
64	SwiFT_NewE96	scratch	zero	mlp	per_subject	0.6215	0.0449	0.6469	0.0392	0.5827	0.0416	25

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#	Feature	Init	Padding	Head	Mode	AUROC		AUPRC		balAcc		n
						μ	σ	μ	σ	μ	σ	
65	NeuroSTORM	scratch	mean	mlp	pooled	0.6193	0.0195	0.6396	0.0226	0.5840	0.0147	5
66	SwiFT_NewE96	scratch	mean	mlp	pooled	0.6187	0.0173	0.6313	0.0149	0.5895	0.0134	10
67	SwiFT_NewE96	resting	cyclic_replicate	mlp	pooled	0.6180	0.0237	0.6376	0.0268	0.5858	0.0239	5
68	SwiFT_NewE96	resting	mean	mlp	pooled	0.6163	0.0119	0.6362	0.0065	0.5903	0.0120	10
69	SwiFT_NewE192	resting	zero	mlp	pooled	0.6156	0.0037	0.6375	0.0063	0.5883	0.0111	5
70	SwiFT_NewE36	resting	zero	mlp	pooled	0.6151	0.0241	0.6439	0.0222	0.5863	0.0122	5
71	SwiFT_NewE96	scratch	spatial_only	mlp	per_subject	0.6135	0.0328	0.6393	0.0359	0.5840	0.0353	25
72	SwiFT_UAH_5M	resting	zero	linear	pooled	0.6128	0.0248	0.6264	0.0347	0.5876	0.0141	5
73	SwiFT_NewE96	scratch	spatial_only	mlp	pooled	0.6113	0.0130	0.6390	0.0150	0.5807	0.0074	5
74	SwiFT_UAH_51M	scratch	zero	mlp	pooled	0.6108	0.0131	0.6337	0.0171	0.5792	0.0086	5
75	SwiFT_NewE36	resting	zero	mlp	per_subject	0.6107	0.0372	0.6497	0.0405	0.5794	0.0265	25
76	SwiFT_NewE96	scratch	replicate	mlp	pooled	0.6095	0.0241	0.6336	0.0243	0.5772	0.0146	5
77	SwiFT_NewE96	scratch	zero	mlp	pooled	0.6070	0.0065	0.6290	0.0067	0.5732	0.0107	5
78	SwiFT_NewE192	scratch	zero	mlp	pooled	0.6051	0.0147	0.6385	0.0183	0.5713	0.0107	5
79	SwiFT_NewE96	resting	spatial_only	mlp	per_subject	0.6037	0.0393	0.6333	0.0345	0.5691	0.0455	25
80	SwiFT_UAH_5M	scratch	zero	mlp	pooled	0.6024	0.0130	0.6208	0.0137	0.5781	0.0124	5
81	SwiFT_NewE192	resting	zero	mlp	per_subject	0.6015	0.0459	0.6311	0.0400	0.5721	0.0439	25
82	SwiFT_NewE96	resting	zero	mlp	per_subject	0.5999	0.0371	0.6280	0.0315	0.5618	0.0340	25
83	SwiFT_NewE96	resting	spatial_only	mlp	pooled	0.5994	0.0156	0.6181	0.0148	0.5619	0.0214	5
84	SwiFT_NewE36	scratch	zero	linear	pooled	0.5990	0.0074	0.6268	0.0059	0.5712	0.0128	5
85	SwiFT_UAH_5M	resting	zero	linear	per_subject	0.5988	0.0496	0.6263	0.0424	0.5737	0.0446	25
86	SwiFT_NewE96	resting	mean	mlp	per_subject	0.5986	0.0345	0.6276	0.0403	0.5614	0.0289	50
87	SwiFT_NewE96	resting	zero	mlp	pooled	0.5949	0.0186	0.6165	0.0125	0.5592	0.0253	5
88	SwiFT_NewE96	resting	cyclic_replicate	mlp	per_subject	0.5899	0.0484	0.6228	0.0437	0.5649	0.0313	25
89	SwiFT_NewE36	scratch	zero	mlp	per_subject	0.5890	0.0522	0.6210	0.0472	0.5635	0.0404	25
90	SwiFT_NewE36	scratch	zero	mlp	pooled	0.5867	0.0074	0.6064	0.0088	0.5604	0.0121	5
91	SwiFT_NewE96	resting	replicate	mlp	pooled	0.5824	0.0235	0.5991	0.0222	0.5524	0.0229	5
92	Brain-JEPA	scratch	zero	mlp	pooled	0.5771	0.0395	0.6009	0.0379	0.5589	0.0312	5
93	Brain-JEPA	scratch	zero	mlp	per_subject	0.5734	0.0615	0.6059	0.0517	0.5430	0.0482	25
94	SwiFT_NewE96	resting	replicate	mlp	per_subject	0.5727	0.0351	0.6007	0.0271	0.5454	0.0362	25
95	SwiFT_UAH_51M	resting	zero	mlp	per_subject	0.5598	0.0477	0.6027	0.0467	0.5154	0.0316	25
96	SwiFT_UAH_5M	resting	zero	mlp	per_subject	0.5536	0.0470	0.5956	0.0444	0.5348	0.0324	25
97	SwiFT_UAH_5M	resting	zero	mlp	pooled	0.5367	0.0112	0.5765	0.0097	0.5157	0.0098	5
98	SwiFT_UAH_202M	resting	zero	mlp	per_subject	0.5281	0.0427	0.5713	0.0368	0.5110	0.0248	25
99	SwiFT_UAH_51M	resting	zero	mlp	pooled	0.5221	0.0186	0.5589	0.0244	0.5009	0.0021	5
100	chance	n/a	n/a	stratified	stim_level	0.5053	0.0329	0.5446	0.0168	0.5053	0.0329	15
101	chance	n/a	n/a	most_frequent	stim_level	0.5000	0.0000	0.5411	0.0020	0.5000	0.0000	5
102	SwiFT_UAH_202M	resting	zero	mlp	pooled	0.4980	0.0504	0.5350	0.0411	0.5024	0.0042	5
<i>Video models (stimulus features)</i>												
1	CLIP_pretrained	n/a	n/a	linear	stim_level	0.9708	0.0112	0.9755	0.0098	0.9113	0.0145	5
2	CLIP_pretrained	n/a	n/a	mlp	stim_level	0.9520	0.0078	0.9568	0.0060	0.8835	0.0170	5
3	DINOv2_pretrained	n/a	n/a	linear	stim_level	0.9135	0.0071	0.9229	0.0088	0.8272	0.0084	5
4	Qwen-VL_caption	n/a	n/a	linear	stim_level	0.9024	0.0165	0.9118	0.0193	0.8189	0.0058	5
5	DINOv2_pretrained	n/a	n/a	mlp	stim_level	0.8814	0.0078	0.8847	0.0188	0.8031	0.0200	5
6	VideoMAE_pretrained	n/a	n/a	linear	stim_level	0.8782	0.0214	0.8935	0.0197	0.8006	0.0167	5
7	Qwen-VL_caption	n/a	n/a	mlp	stim_level	0.8760	0.0168	0.8875	0.0211	0.7970	0.0231	5
8	V-JEPA2_pretrained	n/a	n/a	linear	stim_level	0.8545	0.0215	0.8650	0.0279	0.7813	0.0200	5
9	VideoMAE_pretrained	n/a	n/a	mlp	stim_level	0.8518	0.0244	0.8583	0.0266	0.7682	0.0311	5
10	V-JEPA2_pretrained	n/a	n/a	mlp	stim_level	0.8453	0.0266	0.8562	0.0320	0.7739	0.0269	5
11	V-JEPA2_scratch	n/a	n/a	linear	stim_level	0.6268	0.0359	0.6633	0.0324	0.5807	0.0404	5
12	V-JEPA2_scratch	n/a	n/a	mlp	stim_level	0.6156	0.0332	0.6453	0.0367	0.5721	0.0232	5
13	VideoMAE_scratch	n/a	n/a	mlp	stim_level	0.6108	0.0483	0.6468	0.0387	0.5588	0.0584	5
14	CLIP_scratch	n/a	n/a	linear	stim_level	0.5953	0.0320	0.6246	0.0285	0.5717	0.0160	5
15	CLIP_scratch	n/a	n/a	mlp	stim_level	0.5907	0.0282	0.6300	0.0133	0.5312	0.0206	5
16	VideoMAE_scratch	n/a	n/a	linear	stim_level	0.5884	0.0207	0.6266	0.0164	0.5516	0.0199	5
17	DINOv2_scratch	n/a	n/a	linear	stim_level	0.5809	0.0067	0.6439	0.0203	0.5525	0.0222	5
18	DINOv2_scratch	n/a	n/a	mlp	stim_level	0.5801	0.0443	0.6290	0.0348	0.5286	0.0341	5

2.2 A_binary

Table 3: Full benchmark for A_binary (Arousal binary classification (extreme Q4 vs Q1)). Main metric: AUROC. Cells sorted by main metric within category.

#	Feature	Init	Padding	Head	Mode	AUROC		AUPRC		balAcc		n
						μ	σ	μ	σ	μ	σ	
Brain models (fMRI input)												
1	ROI_Schaefer400Tian50	n/a	time_mean	linear	pooled	0.6784	0.0532	0.7117	0.0447	0.6378	0.0377	5
2	ROI_Schaefer400Tian50	n/a	time_mean	linear	per_subject	0.6694	0.0437	0.7100	0.0389	0.6272	0.0335	25
3	Brain-JEPA	resting	zero	linear	pooled	0.6624	0.0382	0.6969	0.0212	0.6177	0.0289	5
4	ROI_Schaefer400Tian50	n/a	time_mean	mlp	pooled	0.6595	0.0388	0.6948	0.0308	0.6100	0.0331	5
5	Brain-JEPA	resting	zero	mlp	pooled	0.6489	0.0411	0.6908	0.0335	0.6054	0.0326	5
6	Brain-JEPA	resting	zero	linear	per_subject	0.6480	0.0444	0.6933	0.0333	0.6155	0.0458	25
7	ROI_Schaefer400Tian50	n/a	time_mean	mlp	per_subject	0.6388	0.0433	0.6859	0.0348	0.6010	0.0389	25
8	NeuroSTORM	resting	mean	linear	pooled	0.6372	0.0521	0.6868	0.0439	0.5970	0.0345	5
9	NeuroSTORM	resting	mean	linear	per_subject	0.6306	0.0500	0.6833	0.0466	0.5907	0.0321	25
10	Brain-JEPA	scratch	zero	linear	pooled	0.6282	0.0264	0.6609	0.0144	0.6005	0.0220	5
11	Brain-JEPA	resting	zero	mlp	per_subject	0.6209	0.0319	0.6677	0.0346	0.5725	0.0387	25
12	SwiFT_UAH_202M	resting	zero	linear	pooled	0.6206	0.0332	0.6555	0.0357	0.5881	0.0301	5
13	NeuroSTORM	resting	mean	mlp	pooled	0.6198	0.0317	0.6674	0.0242	0.5853	0.0124	5
14	SwiFT_UAH_51M	scratch	zero	linear	per_subject	0.6159	0.0358	0.6621	0.0304	0.5900	0.0353	25
15	SwiFT_NewE36	resting	zero	linear	pooled	0.6130	0.0370	0.6580	0.0249	0.5857	0.0315	5
16	SwiFT_NewE96	resting	spatial_only	linear	pooled	0.6121	0.0373	0.6541	0.0271	0.5900	0.0303	5
17	SwiFT_NewE192	resting	zero	linear	pooled	0.6093	0.0333	0.6505	0.0212	0.5866	0.0282	5
18	SwiFT_NewE96	resting	mean	linear	pooled	0.6067	0.0330	0.6484	0.0268	0.5800	0.0299	10
19	SwiFT_UAH_202M	scratch	zero	linear	per_subject	0.6064	0.0376	0.6563	0.0350	0.5823	0.0362	25
20	SwiFT_NewE96	resting	zero	linear	pooled	0.6055	0.0316	0.6488	0.0150	0.5772	0.0290	5
21	SwiFT_NewE36	resting	zero	linear	per_subject	0.6055	0.0504	0.6536	0.0398	0.5826	0.0374	25
22	NeuroSTORM	resting	mean	mlp	per_subject	0.6049	0.0484	0.6566	0.0449	0.5720	0.0374	25
23	SwiFT_UAH_202M	scratch	zero	linear	pooled	0.6043	0.0279	0.6484	0.0198	0.5823	0.0277	5
24	SwiFT_NewE192	scratch	zero	linear	per_subject	0.6039	0.0373	0.6548	0.0304	0.5712	0.0308	25
25	SwiFT_UAH_5M	scratch	zero	linear	per_subject	0.6015	0.0343	0.6476	0.0251	0.5727	0.0298	25
26	SwiFT_UAH_202M	resting	zero	linear	per_subject	0.6009	0.0478	0.6543	0.0397	0.5783	0.0342	25
27	SwiFT_NewE36	resting	zero	mlp	per_subject	0.6001	0.0379	0.6517	0.0343	0.5583	0.0450	25
28	SwiFT_NewE36	resting	zero	mlp	pooled	0.5995	0.0310	0.6395	0.0222	0.5792	0.0296	5
29	SwiFT_NewE192	resting	zero	linear	per_subject	0.5993	0.0442	0.6461	0.0309	0.5740	0.0395	25
30	SwiFT_NewE192	scratch	zero	linear	pooled	0.5989	0.0224	0.6472	0.0131	0.5746	0.0212	5
31	SwiFT_NewE192	resting	zero	mlp	pooled	0.5981	0.0287	0.6407	0.0225	0.5338	0.0296	5
32	SwiFT_UAH_5M	scratch	zero	linear	pooled	0.5980	0.0284	0.6428	0.0280	0.5762	0.0261	5
33	SwiFT_NewE192	scratch	zero	mlp	per_subject	0.5973	0.0471	0.6477	0.0424	0.5612	0.0350	25
34	SwiFT_NewE96	resting	cyclic_replicate	linear	pooled	0.5972	0.0317	0.6363	0.0274	0.5760	0.0312	5
35	SwiFT_UAH_51M	resting	zero	linear	pooled	0.5945	0.0320	0.6440	0.0369	0.5641	0.0286	5
36	SwiFT_UAH_51M	resting	zero	linear	per_subject	0.5937	0.0363	0.6481	0.0391	0.5657	0.0304	25
37	SwiFT_UAH_51M	scratch	zero	linear	pooled	0.5936	0.0209	0.6409	0.0188	0.5720	0.0183	5
38	Brain-JEPA	scratch	zero	linear	per_subject	0.5913	0.0504	0.6446	0.0437	0.5738	0.0418	25
39	SwiFT_NewE96	resting	spatial_only	linear	per_subject	0.5889	0.0455	0.6415	0.0461	0.5642	0.0392	25
40	SwiFT_NewE96	resting	zero	linear	per_subject	0.5873	0.0518	0.6403	0.0441	0.5655	0.0468	25
41	SwiFT_NewE36	scratch	zero	linear	pooled	0.5848	0.0237	0.6323	0.0247	0.5601	0.0192	5
42	SwiFT_UAH_51M	scratch	zero	mlp	per_subject	0.5847	0.0384	0.6337	0.0342	0.5514	0.0377	25
43	SwiFT_NewE96	resting	zero	mlp	pooled	0.5840	0.0253	0.6345	0.0200	0.5601	0.0243	5
44	SwiFT_UAH_51M	scratch	zero	mlp	pooled	0.5833	0.0151	0.6295	0.0258	0.5621	0.0130	5
45	NeuroSTORM	scratch	mean	linear	per_subject	0.5830	0.0418	0.6375	0.0341	0.5575	0.0316	25
46	SwiFT_NewE96	scratch	zero	linear	per_subject	0.5825	0.0327	0.6377	0.0293	0.5553	0.0255	25
47	SwiFT_UAH_202M	scratch	zero	mlp	pooled	0.5821	0.0171	0.6286	0.0146	0.5531	0.0206	5
48	SwiFT_UAH_202M	scratch	zero	mlp	per_subject	0.5820	0.0434	0.6284	0.0356	0.5438	0.0408	25
49	SwiFT_NewE96	resting	mean	linear	per_subject	0.5818	0.0450	0.6343	0.0451	0.5596	0.0359	50
50	SwiFT_NewE96	scratch	spatial_only	linear	pooled	0.5809	0.0296	0.6301	0.0228	0.5554	0.0266	5
51	NeuroSTORM	scratch	mean	linear	pooled	0.5806	0.0221	0.6244	0.0177	0.5602	0.0189	5
52	SwiFT_UAH_5M	resting	zero	mlp	pooled	0.5804	0.0191	0.6238	0.0233	0.5673	0.0208	5
53	SwiFT_NewE96	scratch	spatial_only	linear	per_subject	0.5798	0.0369	0.6333	0.0332	0.5552	0.0303	25
54	SwiFT_NewE192	scratch	zero	mlp	pooled	0.5798	0.0246	0.6170	0.0214	0.5613	0.0204	5

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#	Feature	Init	Padding	Head	Mode	AUROC		AUPRC		balAcc		n
						μ	σ	μ	σ	μ	σ	
55	SwiFT_NewE96	resting	spatial_only	mlp	pooled	0.5797	0.0268	0.6312	0.0175	0.5592	0.0155	5
56	SwiFT_NewE96	scratch	mean	linear	per_subject	0.5795	0.0352	0.6334	0.0319	0.5559	0.0293	50
57	SwiFT_NewE96	scratch	mean	linear	pooled	0.5791	0.0267	0.6281	0.0189	0.5550	0.0266	10
58	SwiFT_NewE96	resting	mean	mlp	pooled	0.5779	0.0155	0.6241	0.0061	0.5525	0.0169	10
59	SwiFT_UAH_5M	scratch	zero	mlp	pooled	0.5778	0.0331	0.6340	0.0325	0.5386	0.0194	5
60	SwiFT_NewE96	scratch	cyclic_replicate	linear	per_subject	0.5777	0.0374	0.6310	0.0331	0.5535	0.0294	25
61	SwiFT_NewE96	scratch	cyclic_replicate	linear	pooled	0.5771	0.0252	0.6239	0.0205	0.5554	0.0259	5
62	SwiFT_NewE36	scratch	zero	mlp	pooled	0.5760	0.0249	0.6256	0.0215	0.5471	0.0307	5
63	SwiFT_UAH_5M	scratch	zero	mlp	per_subject	0.5758	0.0356	0.6321	0.0280	0.5568	0.0299	25
64	SwiFT_NewE36	scratch	zero	linear	per_subject	0.5753	0.0381	0.6339	0.0306	0.5597	0.0353	25
65	SwiFT_UAH_51M	resting	zero	mlp	pooled	0.5745	0.0207	0.6212	0.0147	0.5126	0.0283	5
66	SwiFT_NewE36	scratch	zero	mlp	per_subject	0.5736	0.0394	0.6299	0.0394	0.5515	0.0332	25
67	SwiFT_NewE96	resting	replicate	linear	pooled	0.5734	0.0248	0.6240	0.0206	0.5504	0.0221	5
68	SwiFT_NewE192	resting	zero	mlp	per_subject	0.5733	0.0403	0.6292	0.0330	0.5349	0.0320	25
69	SwiFT_UAH_51M	resting	zero	mlp	per_subject	0.5727	0.0352	0.6365	0.0158	0.5291	0.0296	25
70	SwiFT_UAH_5M	resting	zero	linear	pooled	0.5708	0.0322	0.6203	0.0296	0.5565	0.0290	5
71	SwiFT_NewE96	scratch	zero	mlp	pooled	0.5703	0.0205	0.6145	0.0193	0.5493	0.0113	5
72	NeuroSTORM	scratch	mean	mlp	pooled	0.5670	0.0236	0.6059	0.0149	0.5503	0.0184	5
73	SwiFT_NewE96	scratch	spatial_only	mlp	pooled	0.5655	0.0210	0.6112	0.0143	0.5398	0.0192	5
74	SwiFT_UAH_5M	resting	zero	linear	per_subject	0.5654	0.0432	0.6207	0.0316	0.5548	0.0330	25
75	SwiFT_NewE96	scratch	zero	mlp	per_subject	0.5650	0.0542	0.6133	0.0500	0.5406	0.0475	25
76	SwiFT_NewE96	resting	cyclic_replicate	linear	per_subject	0.5649	0.0478	0.6195	0.0444	0.5481	0.0355	25
77	SwiFT_NewE96	resting	zero	mlp	per_subject	0.5643	0.0375	0.6261	0.0322	0.5340	0.0250	25
78	SwiFT_NewE96	scratch	zero	linear	pooled	0.5642	0.0191	0.6238	0.0126	0.5397	0.0159	5
79	SwiFT_UAH_5M	resting	zero	mlp	per_subject	0.5631	0.0318	0.6229	0.0331	0.5412	0.0404	25
80	Brain-JEPA	scratch	zero	mlp	pooled	0.5616	0.0164	0.6128	0.0119	0.5417	0.0080	5
81	SwiFT_NewE96	scratch	mean	mlp	per_subject	0.5598	0.0403	0.6200	0.0372	0.5388	0.0322	50
82	SwiFT_NewE96	scratch	mean	mlp	pooled	0.5596	0.0246	0.6095	0.0187	0.5347	0.0221	10
83	SwiFT_NewE96	resting	cyclic_replicate	mlp	pooled	0.5588	0.0177	0.6046	0.0169	0.5286	0.0212	5
84	SwiFT_NewE96	resting	replicate	linear	per_subject	0.5567	0.0408	0.6172	0.0357	0.5417	0.0359	25
85	SwiFT_UAH_202M	resting	zero	mlp	per_subject	0.5565	0.0324	0.6237	0.0238	0.5158	0.0291	25
86	SwiFT_NewE96	scratch	replicate	linear	per_subject	0.5550	0.0486	0.6177	0.0387	0.5370	0.0343	25
87	SwiFT_NewE96	scratch	cyclic_replicate	mlp	per_subject	0.5540	0.0409	0.6161	0.0409	0.5376	0.0334	25
88	SwiFT_UAH_202M	resting	zero	mlp	pooled	0.5538	0.0233	0.6193	0.0198	0.5346	0.0317	5
89	SwiFT_NewE96	scratch	spatial_only	mlp	per_subject	0.5532	0.0472	0.6133	0.0417	0.5345	0.0400	25
90	SwiFT_NewE96	scratch	cyclic_replicate	mlp	pooled	0.5510	0.0158	0.6086	0.0208	0.5338	0.0225	5
91	SwiFT_NewE96	resting	spatial_only	mlp	per_subject	0.5444	0.0493	0.6139	0.0419	0.5259	0.0299	25
92	NeuroSTORM	scratch	mean	mlp	per_subject	0.5441	0.0371	0.6029	0.0372	0.5405	0.0266	25
93	Brain-JEPA	scratch	zero	mlp	per_subject	0.5414	0.0378	0.6076	0.0281	0.5265	0.0252	25
94	SwiFT_NewE96	resting	mean	mlp	per_subject	0.5405	0.0498	0.6170	0.0395	0.5131	0.0305	50
95	SwiFT_NewE96	scratch	replicate	mlp	per_subject	0.5358	0.0504	0.6064	0.0361	0.5218	0.0324	25
96	SwiFT_NewE96	resting	cyclic_replicate	mlp	per_subject	0.5344	0.0502	0.6062	0.0395	0.5215	0.0297	25
97	SwiFT_NewE96	resting	replicate	mlp	pooled	0.5326	0.0173	0.5851	0.0165	0.5202	0.0170	5
98	SwiFT_NewE96	scratch	replicate	linear	pooled	0.5315	0.0233	0.5897	0.0134	0.5217	0.0201	5
99	SwiFT_NewE96	resting	replicate	mlp	per_subject	0.5300	0.0435	0.5970	0.0387	0.5069	0.0312	25
100	SwiFT_NewE96	scratch	replicate	mlp	pooled	0.5166	0.0230	0.5757	0.0147	0.5077	0.0184	5
101	chance	n/a	n/a	most_frequent	stim_level	0.5000	0.0000	0.5619	0.0050	0.5000	0.0000	5
102	chance	n/a	n/a	stratified	stim_level	0.4795	0.0301	0.5528	0.0157	0.4795	0.0301	15
<i>Video models (stimulus features)</i>												
1	CLIP_pretrained	n/a	n/a	linear	stim_level	0.8003	0.0521	0.8232	0.0353	0.7349	0.0559	5
2	CLIP_pretrained	n/a	n/a	mlp	stim_level	0.7813	0.0401	0.7921	0.0455	0.6748	0.1033	5
3	VideoMAE_pretrained	n/a	n/a	linear	stim_level	0.7465	0.0196	0.7869	0.0163	0.6618	0.0098	5
4	V-JEPA2_pretrained	n/a	n/a	linear	stim_level	0.7339	0.0380	0.7581	0.0215	0.6738	0.0521	5
5	DINOv2_pretrained	n/a	n/a	linear	stim_level	0.7278	0.0603	0.7601	0.0465	0.6611	0.0648	5
6	Qwen-VL_caption	n/a	n/a	linear	stim_level	0.7106	0.0261	0.7478	0.0221	0.6556	0.0275	5
7	DINOv2_pretrained	n/a	n/a	mlp	stim_level	0.7071	0.0493	0.7487	0.0405	0.6292	0.0522	5
8	Qwen-VL_caption	n/a	n/a	mlp	stim_level	0.7001	0.0246	0.7371	0.0294	0.6395	0.0357	5
9	V-JEPA2_pretrained	n/a	n/a	mlp	stim_level	0.6940	0.0572	0.7282	0.0422	0.6332	0.0327	5
10	VideoMAE_pretrained	n/a	n/a	mlp	stim_level	0.6893	0.0430	0.7194	0.0492	0.6364	0.0402	5

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#	Feature	Init	Padding	Head	Mode	AUROC		AUPRC		balAcc		n
						μ	σ	μ	σ	μ	σ	
11	VideoMAE_scratch	n/a	n/a	linear	stim_level	0.5946	0.0269	0.6485	0.0226	0.5673	0.0192	5
12	CLIP_scratch	n/a	n/a	mlp	stim_level	0.5804	0.0341	0.6344	0.0369	0.5440	0.0326	5
13	DINOv2_scratch	n/a	n/a	linear	stim_level	0.5740	0.0357	0.6255	0.0513	0.5582	0.0199	5
14	DINOv2_scratch	n/a	n/a	mlp	stim_level	0.5736	0.0439	0.6242	0.0457	0.5531	0.0451	5
15	VideoMAE_scratch	n/a	n/a	mlp	stim_level	0.5733	0.0466	0.6316	0.0437	0.5408	0.0517	5
16	V-JEPA2_scratch	n/a	n/a	linear	stim_level	0.5726	0.0233	0.6262	0.0333	0.5584	0.0171	5
17	V-JEPA2_scratch	n/a	n/a	mlp	stim_level	0.5720	0.0418	0.6167	0.0468	0.5296	0.0205	5
18	CLIP_scratch	n/a	n/a	linear	stim_level	0.5479	0.0434	0.6091	0.0288	0.5491	0.0378	5

2.3 V_reg

Table 4: Full benchmark for V_reg (Valence continuous regression). Main metric: Pearson r. Cells sorted by main metric within category.

#	Feature	Init	Padding	Head	Mode	r		MAE		MSE		RMSE		n
						μ	σ	μ	σ	μ	σ	μ	σ	
Brain models (fMRI input)														
1	ROI_Schaefer400Tian50	n/a	time_mean	mlp	pooled	0.4158	0.0220	1.4870	0.0547	3.4119	0.1817	1.8466	0.0492	5
2	ROI_Schaefer400Tian50	n/a	time_mean	linear	per_subject	0.4042	0.0310	1.4963	0.0326	3.3132	0.1306	1.8199	0.0358	25
3	ROI_Schaefer400Tian50	n/a	time_mean	mlp	per_subject	0.4024	0.0446	1.5619	0.0978	3.8856	0.5979	1.9659	0.1480	25
4	ROI_Schaefer400Tian50	n/a	time_mean	linear	pooled	0.3964	0.0131	1.4991	0.0174	3.2920	0.0421	1.8144	0.0116	5
5	Brain-JEPA	resting	zero	linear	pooled	0.3296	0.0245	1.5539	0.0260	3.4606	0.1331	1.8600	0.0360	5
6	Brain-JEPA	resting	zero	linear	per_subject	0.3275	0.0486	1.5584	0.0443	3.5205	0.2157	1.8755	0.0570	25
7	NeuroSTORM	resting	mean	linear	per_subject	0.3119	0.0471	1.5696	0.0386	3.5814	0.1761	1.8919	0.0465	25
8	Brain-JEPA	resting	zero	mlp	pooled	0.3112	0.0188	1.5640	0.0269	3.6626	0.2273	1.9131	0.0593	5
9	NeuroSTORM	resting	mean	mlp	per_subject	0.2988	0.0519	1.6157	0.0807	4.1203	0.5295	2.0260	0.1275	25
10	NeuroSTORM	resting	mean	mlp	pooled	0.2972	0.0209	1.5866	0.0264	3.8093	0.2562	1.9509	0.0648	5
11	NeuroSTORM	resting	mean	linear	pooled	0.2797	0.0229	1.5809	0.0186	3.5844	0.0951	1.8931	0.0251	5
12	Brain-JEPA	resting	zero	mlp	per_subject	0.2777	0.0575	1.6218	0.0766	3.9945	0.4785	1.9953	0.1181	25
13	SwiFT_NewE96	resting	spatial_only	linear	per_subject	0.2647	0.0408	1.6093	0.0351	3.7721	0.2047	1.9415	0.0525	25
14	SwiFT_NewE96	resting	mean	linear	per_subject	0.2640	0.0397	1.6098	0.0344	3.7751	0.1992	1.9423	0.0511	50
15	SwiFT_UAH_202M	resting	zero	linear	per_subject	0.2636	0.0592	1.6148	0.0313	3.7249	0.2034	1.9293	0.0528	25
16	SwiFT_NewE96	resting	cyclic_replicate	linear	per_subject	0.2618	0.0411	1.6053	0.0333	3.7394	0.1749	1.9333	0.0450	25
17	SwiFT_NewE96	resting	spatial_only	linear	pooled	0.2596	0.0209	1.5970	0.0197	3.6440	0.0925	1.9088	0.0242	5
18	SwiFT_NewE96	resting	zero	linear	per_subject	0.2564	0.0359	1.6176	0.0280	3.7958	0.1771	1.9478	0.0454	25
19	SwiFT_NewE96	resting	mean	linear	pooled	0.2561	0.0141	1.6003	0.0136	3.6672	0.0750	1.9149	0.0196	10
20	SwiFT_NewE96	resting	cyclic_replicate	linear	pooled	0.2559	0.0220	1.5986	0.0186	3.6400	0.0864	1.9078	0.0226	5
21	SwiFT_NewE192	resting	zero	linear	pooled	0.2552	0.0348	1.6000	0.0205	3.6449	0.1297	1.9089	0.0338	5
22	SwiFT_NewE96	resting	zero	linear	pooled	0.2510	0.0204	1.6014	0.0145	3.6557	0.0984	1.9119	0.0259	5
23	SwiFT_UAH_202M	resting	zero	linear	pooled	0.2468	0.0169	1.6112	0.0168	3.7574	0.1663	1.9380	0.0433	5
24	Brain-JEPA	scratch	zero	linear	pooled	0.2440	0.0254	1.6072	0.0288	3.6881	0.1225	1.9202	0.0321	5
25	SwiFT_NewE192	resting	zero	linear	per_subject	0.2340	0.0556	1.6374	0.0551	3.9481	0.3497	1.9852	0.0860	25
26	SwiFT_UAH_51M	resting	zero	linear	per_subject	0.2237	0.0554	1.6580	0.0956	3.9999	0.5420	1.9959	0.1301	25
27	SwiFT_UAH_51M	resting	zero	linear	pooled	0.2200	0.0253	1.6260	0.0235	3.7920	0.1710	1.9469	0.0436	5
28	SwiFT_NewE36	resting	zero	linear	pooled	0.2168	0.0269	1.6194	0.0173	3.7184	0.1190	1.9281	0.0310	5
29	SwiFT_UAH_5M	scratch	zero	linear	per_subject	0.2136	0.0460	1.6501	0.0352	3.9702	0.1591	1.9922	0.0402	25
30	SwiFT_NewE36	resting	zero	linear	per_subject	0.2133	0.0473	1.6319	0.0348	3.8122	0.2065	1.9518	0.0524	25
31	SwiFT_UAH_202M	scratch	zero	mlp	per_subject	0.2085	0.0543	1.7550	0.1148	4.6767	0.7075	2.1568	0.1610	25
32	SwiFT_NewE96	resting	replicate	linear	pooled	0.2042	0.0246	1.6215	0.0177	3.7248	0.0859	1.9299	0.0223	5
33	SwiFT_NewE96	resting	replicate	linear	per_subject	0.1990	0.0443	1.6458	0.0576	3.9236	0.3313	1.9792	0.0807	25
34	SwiFT_NewE96	scratch	zero	linear	per_subject	0.1950	0.0559	1.7352	0.0681	4.5083	0.3186	2.1220	0.0751	25
35	SwiFT_UAH_202M	scratch	zero	mlp	pooled	0.1937	0.0090	1.6646	0.0537	4.0558	0.1832	2.0135	0.0452	5
36	Brain-JEPA	scratch	zero	linear	per_subject	0.1932	0.0415	1.6722	0.0833	4.2973	1.0324	2.0622	0.2155	25
37	SwiFT_NewE96	scratch	spatial_only	mlp	per_subject	0.1926	0.0618	1.7714	0.0893	4.8950	0.6129	2.2082	0.1408	25
38	SwiFT_UAH_51M	scratch	zero	linear	pooled	0.1909	0.0174	1.6511	0.0110	3.9250	0.0994	1.9810	0.0250	5

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#	Feature	Init	Padding	Head	Mode	r		MAE		MSE		RMSE		n
						μ	σ	μ	σ	μ	σ	μ	σ	
39	SwiFT_UAH_5M	scratch	zero	mlp	per_subject	0.1902	0.0531	1.7397	0.1010	4.7242	0.6107	2.1694	0.1372	25
40	SwiFT_NewE96	scratch	cyclic_replicate	linear	per_subject	0.1896	0.0420	1.7597	0.0528	4.6776	0.2429	2.1621	0.0567	25
41	SwiFT_NewE96	scratch	mean	mlp	pooled	0.1890	0.0214	1.6697	0.0385	4.1020	0.3633	2.0236	0.0877	10
42	SwiFT_UAH_51M	scratch	zero	linear	per_subject	0.1882	0.0391	1.7347	0.0547	4.5405	0.2436	2.1301	0.0571	25
43	SwiFT_NewE96	scratch	mean	mlp	per_subject	0.1878	0.0592	1.7837	0.1147	4.9446	0.7483	2.2174	0.1686	50
44	SwiFT_NewE96	scratch	mean	linear	per_subject	0.1867	0.0480	1.7706	0.0504	4.7433	0.2597	2.1771	0.0602	50
45	SwiFT_NewE192	scratch	zero	mlp	per_subject	0.1842	0.0426	1.7988	0.1605	4.9706	0.8829	2.2213	0.1943	25
46	SwiFT_NewE96	scratch	zero	linear	pooled	0.1835	0.0184	1.6447	0.0187	3.9187	0.0959	1.9795	0.0242	5
47	NeuroSTORM	scratch	mean	linear	pooled	0.1834	0.0310	1.6355	0.0238	3.7948	0.0959	1.9479	0.0245	5
48	SwiFT_NewE96	scratch	spatial_only	linear	per_subject	0.1830	0.0523	1.7754	0.0543	4.7670	0.2774	2.1824	0.0643	25
49	SwiFT_NewE96	scratch	cyclic_replicate	mlp	per_subject	0.1830	0.0461	1.8322	0.2253	5.2579	1.4613	2.2776	0.2708	25
50	SwiFT_NewE96	scratch	cyclic_replicate	linear	pooled	0.1828	0.0134	1.6499	0.0092	3.9505	0.0442	1.9876	0.0111	5
51	SwiFT_NewE96	scratch	zero	mlp	per_subject	0.1816	0.0525	1.8608	0.3291	5.3687	2.0909	2.2882	0.3717	25
52	NeuroSTORM	scratch	mean	mlp	per_subject	0.1797	0.0550	1.7770	0.1385	4.8960	0.8880	2.2041	0.1983	25
53	SwiFT_NewE96	scratch	mean	linear	pooled	0.1795	0.0134	1.6532	0.0059	3.9627	0.0407	1.9906	0.0102	10
54	NeuroSTORM	scratch	mean	mlp	pooled	0.1779	0.0272	1.6544	0.0275	3.9036	0.1312	1.9755	0.0333	5
55	SwiFT_UAH_202M	scratch	zero	linear	per_subject	0.1778	0.0435	1.8657	0.0623	5.3343	0.3369	2.3085	0.0726	25
56	SwiFT_NewE96	resting	mean	mlp	pooled	0.1772	0.0172	1.6625	0.0288	4.0598	0.1831	2.0144	0.0460	10
57	SwiFT_UAH_51M	scratch	zero	mlp	pooled	0.1769	0.0132	1.6549	0.0335	4.0195	0.2556	2.0041	0.0630	5
58	SwiFT_NewE96	scratch	spatial_only	mlp	pooled	0.1762	0.0161	1.6727	0.0181	3.9965	0.1345	1.9989	0.0337	5
59	SwiFT_UAH_51M	scratch	zero	mlp	per_subject	0.1760	0.0394	1.7701	0.1134	4.7604	0.7542	2.1753	0.1720	25
60	SwiFT_NewE96	scratch	replicate	mlp	per_subject	0.1748	0.0426	1.8202	0.1207	4.9307	0.7766	2.2139	0.1750	25
61	NeuroSTORM	scratch	mean	linear	per_subject	0.1744	0.0637	1.6897	0.0600	4.1743	0.2999	2.0419	0.0715	25
62	SwiFT_NewE96	scratch	spatial_only	linear	pooled	0.1729	0.0145	1.6572	0.0051	3.9778	0.0392	1.9944	0.0098	5
63	SwiFT_UAH_202M	scratch	zero	linear	pooled	0.1729	0.0131	1.7033	0.0158	4.2536	0.0958	2.0623	0.0232	5
64	SwiFT_UAH_5M	scratch	zero	linear	pooled	0.1693	0.0229	1.6419	0.0116	3.8180	0.0674	1.9539	0.0172	5
65	SwiFT_NewE96	scratch	cyclic_replicate	mlp	pooled	0.1693	0.0124	1.6883	0.0739	4.2493	0.5169	2.0584	0.1237	5
66	SwiFT_NewE36	scratch	zero	linear	per_subject	0.1686	0.0600	1.6831	0.0591	4.1106	0.2871	2.0263	0.0702	25
67	SwiFT_NewE96	resting	cyclic_replicate	mlp	pooled	0.1672	0.0160	1.6531	0.0359	3.9118	0.1526	1.9775	0.0386	5
68	SwiFT_NewE96	scratch	replicate	mlp	pooled	0.1646	0.0159	1.7162	0.0836	4.3549	0.4910	2.0842	0.1169	5
69	SwiFT_NewE96	resting	spatial_only	mlp	pooled	0.1625	0.0145	1.6561	0.0272	4.0419	0.2574	2.0096	0.0635	5
70	SwiFT_UAH_5M	scratch	zero	mlp	pooled	0.1554	0.0213	1.7066	0.0875	4.4304	0.6049	2.1010	0.1418	5
71	SwiFT_UAH_5M	resting	zero	linear	per_subject	0.1544	0.0483	1.6817	0.0371	4.0770	0.2867	2.0180	0.0697	25
72	SwiFT_NewE96	scratch	replicate	linear	pooled	0.1541	0.0213	1.6654	0.0093	4.0109	0.0789	2.0027	0.0197	5
73	SwiFT_NewE192	scratch	zero	linear	per_subject	0.1526	0.0436	1.9192	0.0595	5.6023	0.3324	2.3659	0.0693	25
74	SwiFT_NewE36	resting	zero	mlp	pooled	0.1525	0.0224	1.6622	0.0130	3.9765	0.0919	1.9940	0.0229	5
75	SwiFT_NewE96	scratch	zero	mlp	pooled	0.1514	0.0176	1.6710	0.0391	4.1765	0.2964	2.0426	0.0727	5
76	SwiFT_NewE96	scratch	replicate	linear	per_subject	0.1508	0.0523	1.7998	0.0597	4.8889	0.2985	2.2101	0.0678	25
77	SwiFT_UAH_5M	resting	zero	linear	pooled	0.1464	0.0382	1.6596	0.0300	3.8908	0.1964	1.9720	0.0494	5
78	SwiFT_NewE192	scratch	zero	linear	pooled	0.1449	0.0094	1.7144	0.0184	4.3561	0.0713	2.0871	0.0171	5
79	SwiFT_NewE36	resting	zero	mlp	per_subject	0.1439	0.0542	1.6865	0.0578	4.1759	0.3465	2.0418	0.0842	25
80	SwiFT_NewE192	resting	zero	mlp	per_subject	0.1433	0.0797	1.7811	0.1087	4.7514	0.6562	2.1747	0.1511	25
81	SwiFT_NewE192	scratch	zero	mlp	pooled	0.1430	0.0056	1.7243	0.0714	4.5077	0.3640	2.1217	0.0865	5
82	SwiFT_NewE192	resting	zero	mlp	pooled	0.1420	0.0089	1.7210	0.0459	4.3376	0.3251	2.0815	0.0778	5
83	SwiFT_NewE36	scratch	zero	linear	pooled	0.1408	0.0112	1.6522	0.0132	3.8467	0.0741	1.9612	0.0190	5
84	SwiFT_NewE96	resting	spatial_only	mlp	per_subject	0.1392	0.0551	1.7272	0.0786	4.3532	0.5080	2.0832	0.1184	25
85	SwiFT_NewE96	resting	zero	mlp	per_subject	0.1387	0.0416	1.6800	0.0578	4.0998	0.3376	2.0232	0.0824	25
86	SwiFT_NewE96	resting	zero	mlp	pooled	0.1346	0.0255	1.6620	0.0428	3.9945	0.1304	1.9984	0.0327	5
87	SwiFT_NewE96	resting	cyclic_replicate	mlp	per_subject	0.1340	0.0431	1.6833	0.0594	4.1413	0.3359	2.0335	0.0814	25
88	SwiFT_NewE96	resting	mean	mlp	per_subject	0.1306	0.0377	1.7283	0.0681	4.3470	0.3870	2.0830	0.0910	50
89	SwiFT_NewE96	resting	replicate	mlp	pooled	0.1221	0.0317	1.6752	0.0454	3.9742	0.1911	1.9931	0.0480	5
90	SwiFT_NewE36	scratch	zero	mlp	per_subject	0.1216	0.0632	1.7616	0.1308	4.7052	0.8480	2.1612	0.1892	25
91	SwiFT_NewE36	scratch	zero	mlp	pooled	0.1187	0.0072	1.6512	0.0158	3.9639	0.0571	1.9909	0.0143	5
92	Brain-JEPA	scratch	zero	mlp	per_subject	0.1016	0.0619	1.7281	0.1586	4.2628	0.7745	2.0582	0.1665	25
93	Brain-JEPA	scratch	zero	mlp	pooled	0.0931	0.0339	1.6592	0.0196	3.8961	0.0403	1.9738	0.0102	5
94	SwiFT_NewE96	resting	replicate	mlp	per_subject	0.0798	0.0703	1.7325	0.0809	4.3634	0.5136	2.0856	0.1198	25
95	SwiFT_UAH_5M	resting	zero	mlp	per_subject	0.0612	0.0616	1.6800	0.0323	3.9327	0.1089	1.9829	0.0274	25
96	SwiFT_UAH_51M	resting	zero	mlp	per_subject	0.0339	0.0722	1.7582	0.1992	4.4287	1.3037	2.0874	0.2728	25
97	SwiFT_UAH_5M	resting	zero	mlp	pooled	0.0267	0.0312	1.6768	0.0132	3.8925	0.0852	1.9728	0.0217	5
98	SwiFT_UAH_51M	resting	zero	mlp	pooled	0.0199	0.0354	1.7639	0.1852	4.2682	0.8342	2.0589	0.1908	5

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#	Feature	Init	Padding	Head	Mode	r		MAE		MSE		RMSE		n
						μ	σ	μ	σ	μ	σ	μ	σ	
99	SwiFT_UAH_202M	resting	zero	mlp	per_subject	0.0107	0.0460	1.7729	0.1653	4.6213	0.9666	2.1397	0.2114	25
100	chance	n/a	n/a	mean	stim_level	0.0000	0.0000	1.6711	0.0125	3.8599	0.0713	1.9646	0.0182	5
101	chance	n/a	n/a	median	stim_level	0.0000	0.0000	1.6392	0.0107	4.0468	0.0860	2.0116	0.0214	5
102	SwiFT_UAH_202M	resting	zero	mlp	pooled	-0.0081	0.0539	1.6626	0.0281	4.0175	0.1979	2.0039	0.0492	5
<i>Video models (stimulus features)</i>														
1	CLIP_pretrained	n/a	n/a	mlp	stim_level	0.7645	0.0309	1.0446	0.0647	1.8286	0.3121	1.3484	0.1143	5
2	CLIP_pretrained	n/a	n/a	linear	stim_level	0.6832	0.0243	1.2011	0.0386	2.3084	0.1712	1.5185	0.0559	5
3	DINOv2_pretrained	n/a	n/a	mlp	stim_level	0.6008	0.0360	1.3958	0.1378	3.1539	0.5791	1.7698	0.1649	5
4	Qwen-VL_caption	n/a	n/a	mlp	stim_level	0.5990	0.0395	1.3723	0.1971	3.1159	1.0313	1.7485	0.2708	5
5	Qwen-VL_caption	n/a	n/a	linear	stim_level	0.5741	0.0263	1.3189	0.0505	2.7260	0.1881	1.6503	0.0566	5
6	VideoMAE_pretrained	n/a	n/a	mlp	stim_level	0.5455	0.0351	1.5762	0.1901	3.9391	0.8208	1.9758	0.2104	5
7	V-JEPA2_pretrained	n/a	n/a	mlp	stim_level	0.5296	0.0213	1.4617	0.0960	3.4231	0.5364	1.8456	0.1449	5
8	VideoMAE_pretrained	n/a	n/a	linear	stim_level	0.5201	0.0399	1.4179	0.0550	3.1629	0.2603	1.7772	0.0733	5
9	DINOv2_pretrained	n/a	n/a	linear	stim_level	0.4964	0.0138	1.5299	0.0488	3.7741	0.1781	1.9423	0.0463	5
10	V-JEPA2_pretrained	n/a	n/a	linear	stim_level	0.4703	0.0394	1.5205	0.0701	3.5625	0.3103	1.8860	0.0826	5
11	V-JEPA2_scratch	n/a	n/a	linear	stim_level	0.1459	0.0305	1.6789	0.0131	4.0101	0.1000	2.0024	0.0249	5
12	V-JEPA2_scratch	n/a	n/a	mlp	stim_level	0.1381	0.0121	1.6977	0.0506	4.0933	0.3104	2.0220	0.0766	5
13	VideoMAE_scratch	n/a	n/a	linear	stim_level	0.1379	0.0092	1.6786	0.0289	3.9706	0.1564	1.9923	0.0388	5
14	CLIP_scratch	n/a	n/a	linear	stim_level	0.1345	0.0444	1.6543	0.0313	3.9014	0.1608	1.9749	0.0404	5
15	VideoMAE_scratch	n/a	n/a	mlp	stim_level	0.1123	0.0367	1.6900	0.0516	4.0724	0.2418	2.0173	0.0593	5
16	CLIP_scratch	n/a	n/a	mlp	stim_level	0.1011	0.0308	1.6744	0.0553	4.1187	0.2291	2.0288	0.0567	5
17	DINOv2_scratch	n/a	n/a	mlp	stim_level	0.0934	0.0337	1.8569	0.1074	4.8862	0.4544	2.2086	0.1020	5
18	DINOv2_scratch	n/a	n/a	linear	stim_level	0.0696	0.0401	3.3389	3.3599	38.0838	73.9443	4.4015	4.8362	5

2.4 A_reg

Table 5: Full benchmark for A_reg (Arousal continuous regression). Main metric: Pearson r. Cells sorted by main metric within category.

#	Feature	Init	Padding	Head	Mode	r		MAE		MSE		RMSE		n
						μ	σ	μ	σ	μ	σ	μ	σ	
<i>Brain models (fMRI input)</i>														
1	ROI_Schaefer400Tian50	n/a	time_mean	linear	pooled	0.2334	0.0524	0.7120	0.0090	0.8061	0.0256	0.8978	0.0142	5
2	ROI_Schaefer400Tian50	n/a	time_mean	mlp	pooled	0.2254	0.0497	0.7328	0.0266	0.8544	0.0716	0.9237	0.0382	5
3	Brain-JEPA	resting	zero	linear	pooled	0.2214	0.0302	0.7071	0.0105	0.7964	0.0365	0.8922	0.0206	5
4	ROI_Schaefer400Tian50	n/a	time_mean	linear	per_subject	0.2207	0.0576	0.7232	0.0196	0.8364	0.0443	0.9142	0.0240	25
5	Brain-JEPA	resting	zero	mlp	pooled	0.2138	0.0389	0.7211	0.0151	0.8290	0.0489	0.9102	0.0269	5
6	Brain-JEPA	resting	zero	linear	per_subject	0.2011	0.0484	0.7201	0.0222	0.8322	0.0735	0.9115	0.0389	25
7	ROI_Schaefer400Tian50	n/a	time_mean	mlp	per_subject	0.2003	0.0567	0.7641	0.0543	0.9426	0.1393	0.9684	0.0704	25
8	NeuroSTORM	resting	mean	linear	pooled	0.1911	0.0600	0.7134	0.0070	0.8114	0.0252	0.9007	0.0140	5
9	Brain-JEPA	resting	zero	mlp	per_subject	0.1737	0.0525	0.7378	0.0235	0.8634	0.0653	0.9285	0.0349	25
10	SwiFT_NewE96	resting	spatial_only	linear	pooled	0.1591	0.0440	0.7145	0.0082	0.8151	0.0297	0.9027	0.0166	5
11	Brain-JEPA	scratch	zero	linear	pooled	0.1583	0.0269	0.7202	0.0099	0.8279	0.0346	0.9097	0.0192	5
12	NeuroSTORM	resting	mean	mlp	pooled	0.1575	0.0455	0.7258	0.0177	0.8309	0.0414	0.9113	0.0230	5
13	SwiFT_NewE36	resting	zero	mlp	pooled	0.1560	0.0523	0.7178	0.0069	0.8187	0.0213	0.9047	0.0118	5
14	SwiFT_NewE96	resting	mean	linear	pooled	0.1538	0.0420	0.7153	0.0075	0.8168	0.0281	0.9037	0.0157	10
15	SwiFT_NewE36	resting	zero	linear	pooled	0.1538	0.0382	0.7184	0.0077	0.8158	0.0277	0.9031	0.0154	5
16	SwiFT_UAH_202M	scratch	zero	mlp	pooled	0.1456	0.0323	0.7818	0.0586	0.9746	0.1218	0.9857	0.0620	5
17	SwiFT_UAH_202M	resting	zero	linear	pooled	0.1453	0.0351	0.7205	0.0086	0.8235	0.0293	0.9074	0.0161	5
18	SwiFT_NewE96	resting	cyclic_replicate	linear	pooled	0.1447	0.0434	0.7162	0.0080	0.8190	0.0304	0.9048	0.0169	5
19	SwiFT_NewE192	resting	zero	linear	pooled	0.1432	0.0304	0.7227	0.0094	0.8328	0.0314	0.9125	0.0173	5
20	SwiFT_UAH_51M	resting	zero	linear	pooled	0.1414	0.0309	0.7194	0.0090	0.8198	0.0297	0.9053	0.0164	5
21	SwiFT_NewE96	resting	zero	linear	pooled	0.1414	0.0287	0.7171	0.0089	0.8185	0.0334	0.9046	0.0186	5
22	SwiFT_UAH_202M	resting	zero	linear	per_subject	0.1407	0.0564	0.7386	0.0673	0.8706	0.1789	0.9293	0.0850	25
23	SwiFT_NewE192	scratch	zero	mlp	pooled	0.1385	0.0169	0.7464	0.0291	0.8897	0.0808	0.9424	0.0432	5

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#	Feature	Init	Padding	Head	Mode	r		MAE		MSE		RMSE		n
						μ	σ	μ	σ	μ	σ	μ	σ	
24	SwiFT_UAH_51M	resting	zero	linear	per_subject	0.1366	0.0407	0.7276	0.0169	0.8373	0.0441	0.9148	0.0238	25
25	SwiFT_NewE192	resting	zero	mlp	pooled	0.1366	0.0450	0.7198	0.0145	0.8242	0.0412	0.9076	0.0227	5
26	SwiFT_NewE36	resting	zero	linear	per_subject	0.1355	0.0612	0.7358	0.0286	0.8590	0.0691	0.9261	0.0363	25
27	NeuroSTORM	resting	mean	linear	per_subject	0.1340	0.0734	0.7454	0.0338	0.8888	0.0795	0.9419	0.0412	25
28	SwiFT_UAH_51M	scratch	zero	mlp	per_subject	0.1307	0.0442	0.7836	0.0726	0.9881	0.1945	0.9899	0.0927	25
29	SwiFT_UAH_202M	scratch	zero	mlp	per_subject	0.1282	0.0459	0.8512	0.1345	1.1695	0.3610	1.0706	0.1554	25
30	SwiFT_NewE36	resting	zero	mlp	per_subject	0.1257	0.0545	0.7525	0.0327	0.9008	0.0770	0.9483	0.0397	25
31	SwiFT_NewE192	scratch	zero	mlp	per_subject	0.1246	0.0538	0.8206	0.0624	1.0726	0.1648	1.0327	0.0795	25
32	SwiFT_UAH_5M	resting	zero	mlp	pooled	0.1220	0.0323	0.7226	0.0110	0.8265	0.0362	0.9090	0.0198	5
33	SwiFT_NewE96	resting	spatial_only	linear	per_subject	0.1217	0.0597	0.7480	0.0811	0.9025	0.2469	0.9443	0.1058	25
34	SwiFT_UAH_5M	scratch	zero	mlp	pooled	0.1213	0.0326	0.7227	0.0074	0.8295	0.0356	0.9106	0.0196	5
35	SwiFT_NewE96	resting	zero	mlp	pooled	0.1203	0.0292	0.7214	0.0091	0.8256	0.0360	0.9085	0.0199	5
36	NeuroSTORM	resting	mean	mlp	per_subject	0.1199	0.0615	0.7645	0.0587	0.9342	0.1559	0.9637	0.0750	25
37	SwiFT_NewE192	resting	zero	linear	per_subject	0.1197	0.0635	0.7525	0.0362	0.9004	0.0889	0.9478	0.0461	25
38	SwiFT_NewE96	resting	mean	linear	per_subject	0.1166	0.0582	0.7468	0.0736	0.8979	0.2162	0.9429	0.0950	50
39	SwiFT_UAH_51M	scratch	zero	mlp	pooled	0.1147	0.0184	0.7281	0.0077	0.8399	0.0281	0.9164	0.0153	5
40	SwiFT_NewE96	scratch	zero	mlp	per_subject	0.1137	0.0499	0.7863	0.0633	0.9893	0.1723	0.9913	0.0835	25
41	NeuroSTORM	scratch	mean	linear	pooled	0.1132	0.0136	0.7266	0.0097	0.8408	0.0367	0.9168	0.0201	5
42	Brain-JEPA	scratch	zero	linear	per_subject	0.1131	0.0614	0.7388	0.0355	0.9230	0.2184	0.9555	0.1025	25
43	SwiFT_NewE96	resting	spatial_only	mlp	pooled	0.1128	0.0267	0.7243	0.0095	0.8311	0.0297	0.9115	0.0163	5
44	SwiFT_NewE96	resting	replicate	linear	pooled	0.1123	0.0292	0.7221	0.0055	0.8296	0.0237	0.9108	0.0130	5
45	SwiFT_UAH_5M	scratch	zero	mlp	per_subject	0.1114	0.0503	0.7667	0.0519	0.9394	0.1301	0.9671	0.0656	25
46	NeuroSTORM	scratch	mean	mlp	pooled	0.1109	0.0219	0.7303	0.0202	0.8427	0.0619	0.9175	0.0334	5
47	SwiFT_NewE36	scratch	zero	linear	pooled	0.1075	0.0215	0.7276	0.0118	0.8401	0.0345	0.9164	0.0188	5
48	SwiFT_NewE96	resting	cyclic_replicate	linear	per_subject	0.1073	0.0542	0.7509	0.0732	0.9079	0.1961	0.9490	0.0877	25
49	SwiFT_NewE96	resting	zero	linear	per_subject	0.1071	0.0529	0.7729	0.1031	0.9767	0.3201	0.9786	0.1403	25
50	SwiFT_NewE36	scratch	zero	mlp	pooled	0.1061	0.0175	0.7254	0.0092	0.8320	0.0362	0.9120	0.0199	5
51	SwiFT_NewE96	resting	mean	mlp	pooled	0.1059	0.0205	0.7300	0.0108	0.8459	0.0398	0.9195	0.0219	10
52	SwiFT_UAH_5M	resting	zero	linear	pooled	0.1022	0.0278	0.7236	0.0063	0.8304	0.0274	0.9112	0.0151	5
53	SwiFT_NewE192	resting	zero	mlp	per_subject	0.1020	0.0683	0.7727	0.0586	0.9555	0.1607	0.9748	0.0745	25
54	SwiFT_UAH_5M	resting	zero	mlp	per_subject	0.1018	0.0506	0.7292	0.0117	0.8435	0.0311	0.9183	0.0170	25
55	SwiFT_UAH_51M	resting	zero	mlp	pooled	0.1010	0.0178	0.7272	0.0111	0.8397	0.0384	0.9161	0.0211	5
56	SwiFT_UAH_5M	scratch	zero	linear	pooled	0.0997	0.0387	0.7312	0.0082	0.8477	0.0296	0.9206	0.0161	5
57	SwiFT_NewE96	scratch	zero	mlp	pooled	0.0994	0.0202	0.7587	0.0084	0.9122	0.0444	0.9549	0.0229	5
58	SwiFT_UAH_51M	resting	zero	mlp	per_subject	0.0965	0.0269	0.7354	0.0273	0.8586	0.0637	0.9260	0.0334	25
59	Brain-JEPA	scratch	zero	mlp	pooled	0.0962	0.0246	0.7249	0.0082	0.8311	0.0258	0.9116	0.0141	5
60	SwiFT_UAH_51M	scratch	zero	linear	pooled	0.0959	0.0019	0.7480	0.0124	0.8869	0.0417	0.9415	0.0222	5
61	SwiFT_UAH_202M	resting	zero	mlp	per_subject	0.0959	0.0334	0.7457	0.0366	0.8782	0.0828	0.9362	0.0428	25
62	SwiFT_UAH_202M	scratch	zero	linear	pooled	0.0958	0.0230	0.7807	0.0146	0.9620	0.0341	0.9807	0.0174	5
63	NeuroSTORM	scratch	mean	linear	per_subject	0.0950	0.0476	0.7766	0.0259	0.9544	0.0579	0.9765	0.0295	25
64	SwiFT_NewE36	scratch	zero	mlp	per_subject	0.0944	0.0412	0.7874	0.0686	0.9887	0.1798	0.9908	0.0862	25
65	SwiFT_UAH_202M	resting	zero	mlp	pooled	0.0942	0.0269	0.7278	0.0125	0.8371	0.0409	0.9147	0.0223	5
66	SwiFT_UAH_5M	scratch	zero	linear	per_subject	0.0937	0.0459	0.7688	0.0219	0.9395	0.0555	0.9689	0.0285	25
67	SwiFT_NewE36	scratch	zero	linear	per_subject	0.0934	0.0473	0.7682	0.0292	0.9348	0.0650	0.9663	0.0331	25
68	SwiFT_NewE192	scratch	zero	linear	pooled	0.0921	0.0171	0.7822	0.0129	0.9668	0.0336	0.9832	0.0172	5
69	SwiFT_NewE96	resting	zero	mlp	per_subject	0.0917	0.0356	0.7454	0.0232	0.8863	0.0730	0.9407	0.0380	25
70	SwiFT_NewE96	resting	cyclic_replicate	mlp	pooled	0.0914	0.0167	0.7250	0.0071	0.8352	0.0275	0.9138	0.0150	5
71	SwiFT_NewE96	scratch	mean	mlp	pooled	0.0912	0.0146	0.7353	0.0133	0.8660	0.0499	0.9303	0.0264	10
72	SwiFT_NewE96	scratch	zero	linear	pooled	0.0900	0.0196	0.7489	0.0147	0.8887	0.0383	0.9426	0.0203	5
73	SwiFT_NewE96	scratch	spatial_only	linear	pooled	0.0892	0.0267	0.7502	0.0087	0.8909	0.0313	0.9437	0.0166	5
74	SwiFT_NewE96	scratch	cyclic_replicate	mlp	per_subject	0.0891	0.0437	0.8399	0.0914	1.1212	0.2412	1.0536	0.1074	25
75	SwiFT_NewE96	scratch	spatial_only	mlp	pooled	0.0884	0.0369	0.7798	0.0569	0.9745	0.1691	0.9842	0.0846	5
76	SwiFT_NewE96	scratch	mean	linear	pooled	0.0873	0.0249	0.7503	0.0075	0.8920	0.0289	0.9443	0.0154	10
77	SwiFT_NewE96	scratch	spatial_only	mlp	per_subject	0.0855	0.0391	0.8232	0.0818	1.0707	0.2033	1.0305	0.0957	25
78	SwiFT_NewE96	scratch	cyclic_replicate	mlp	pooled	0.0845	0.0184	0.7399	0.0243	0.8746	0.0557	0.9348	0.0295	5
79	SwiFT_NewE96	scratch	cyclic_replicate	linear	pooled	0.0814	0.0211	0.7499	0.0084	0.8938	0.0319	0.9453	0.0169	5
80	SwiFT_UAH_5M	resting	zero	linear	per_subject	0.0804	0.0493	0.7466	0.0307	0.8878	0.0811	0.9413	0.0422	25
81	SwiFT_NewE96	resting	replicate	linear	per_subject	0.0791	0.0590	0.7789	0.0991	0.9819	0.2848	0.9834	0.1245	25
82	NeuroSTORM	scratch	mean	mlp	per_subject	0.0783	0.0684	0.8014	0.0532	1.0264	0.1451	1.0107	0.0709	25
83	SwiFT_UAH_51M	scratch	zero	linear	per_subject	0.0745	0.0490	0.8406	0.0780	1.1260	0.2416	1.0568	0.0978	25

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#	Feature	Init	Padding	Head	Mode	r		MAE		MSE		RMSE		n
						μ	σ	μ	σ	μ	σ	μ	σ	
84	SwiFT_NewE96	resting	spatial_only	mlp	per_subject	0.0718	0.0611	0.7520	0.0275	0.9051	0.0843	0.9504	0.0433	25
85	SwiFT_NewE96	scratch	mean	mlp	per_subject	0.0711	0.0541	0.8299	0.0652	1.0859	0.1672	1.0390	0.0811	50
86	SwiFT_UAH_202M	scratch	zero	linear	per_subject	0.0697	0.0479	1.0735	0.4061	2.1400	1.9995	1.3660	0.5342	25
87	SwiFT_NewE96	scratch	zero	linear	per_subject	0.0690	0.0621	0.8501	0.0903	1.1655	0.2546	1.0744	0.1080	25
88	SwiFT_NewE96	scratch	mean	linear	per_subject	0.0672	0.0383	0.8663	0.0702	1.1960	0.2131	1.0900	0.0897	50
89	SwiFT_NewE96	scratch	replicate	mlp	per_subject	0.0670	0.0556	0.8141	0.0680	1.0491	0.1761	1.0208	0.0854	25
90	SwiFT_NewE96	scratch	spatial_only	linear	per_subject	0.0656	0.0435	0.8775	0.0916	1.2348	0.2830	1.1053	0.1171	25
91	SwiFT_NewE96	scratch	cyclic_replicate	linear	per_subject	0.0651	0.0391	0.8803	0.0906	1.2367	0.2731	1.1065	0.1131	25
92	SwiFT_NewE192	scratch	zero	linear	per_subject	0.0611	0.0525	1.0831	0.3700	2.0331	1.6949	1.3518	0.4630	25
93	SwiFT_NewE96	scratch	replicate	linear	per_subject	0.0591	0.0495	0.9021	0.0937	1.3050	0.2959	1.1360	0.1227	25
94	Brain-JEPA	scratch	zero	mlp	per_subject	0.0559	0.0368	0.7390	0.0168	0.8670	0.0426	0.9308	0.0230	25
95	SwiFT_NewE96	resting	replicate	mlp	pooled	0.0535	0.0237	0.7247	0.0111	0.8313	0.0382	0.9116	0.0210	5
96	SwiFT_NewE96	resting	mean	mlp	per_subject	0.0490	0.0440	0.7509	0.0301	0.8991	0.0920	0.9470	0.0474	50
97	SwiFT_NewE96	scratch	replicate	mlp	pooled	0.0482	0.0187	0.7399	0.0288	0.8689	0.0739	0.9315	0.0397	5
98	SwiFT_NewE96	scratch	replicate	linear	pooled	0.0427	0.0198	0.7559	0.0111	0.9088	0.0378	0.9531	0.0199	5
99	SwiFT_NewE96	resting	cyclic_replicate	mlp	per_subject	0.0392	0.0530	0.7583	0.0363	0.9146	0.0954	0.9551	0.0489	25
100	SwiFT_NewE96	resting	replicate	mlp	per_subject	0.0349	0.0591	0.7826	0.0824	0.9720	0.2014	0.9814	0.0957	25
101	chance	n/a	n/a	mean	stim_level	0.0000	0.0000	0.7253	0.0113	0.8315	0.0385	0.9117	0.0212	5
102	chance	n/a	n/a	median	stim_level	0.0000	0.0000	0.7250	0.0116	0.8358	0.0396	0.9140	0.0217	5
<i>Video models (stimulus features)</i>														
1	CLIP_pretrained	n/a	n/a	mlp	stim_level	0.4231	0.0421	0.7133	0.0391	0.7998	0.0842	0.8933	0.0474	5
2	CLIP_pretrained	n/a	n/a	linear	stim_level	0.3375	0.0472	0.7787	0.0261	0.9634	0.0602	0.9812	0.0305	5
3	V-JEPA2_pretrained	n/a	n/a	mlp	stim_level	0.3237	0.0567	0.8019	0.1029	1.0246	0.2520	1.0065	0.1205	5
4	VideoMAE_pretrained	n/a	n/a	mlp	stim_level	0.3043	0.0307	0.7981	0.1066	1.0090	0.2171	1.0001	0.1048	5
5	Qwen-VL_caption	n/a	n/a	mlp	stim_level	0.3013	0.0287	0.7193	0.0399	0.8447	0.0864	0.9182	0.0459	5
6	DINOv2_pretrained	n/a	n/a	mlp	stim_level	0.2985	0.0453	0.7734	0.0469	0.9663	0.1204	0.9814	0.0636	5
7	VideoMAE_pretrained	n/a	n/a	linear	stim_level	0.2758	0.0417	0.7649	0.0189	0.9420	0.0580	0.9702	0.0299	5
8	V-JEPA2_pretrained	n/a	n/a	linear	stim_level	0.2499	0.0376	0.7940	0.0253	1.0065	0.0639	1.0029	0.0320	5
9	Qwen-VL_caption	n/a	n/a	linear	stim_level	0.2339	0.0626	0.7528	0.0212	0.9016	0.0609	0.9491	0.0327	5
10	DINOv2_pretrained	n/a	n/a	linear	stim_level	0.1864	0.0381	0.9137	0.0167	1.3367	0.0616	1.1559	0.0266	5
11	VideoMAE_scratch	n/a	n/a	linear	stim_level	0.1355	0.0858	0.7400	0.0322	0.8622	0.0634	0.9281	0.0336	5
12	V-JEPA2_scratch	n/a	n/a	mlp	stim_level	0.1112	0.0458	0.7459	0.0217	0.8724	0.0514	0.9337	0.0277	5
13	V-JEPA2_scratch	n/a	n/a	linear	stim_level	0.1107	0.0261	0.8395	0.2075	1.2137	0.7285	1.0711	0.2882	5
14	VideoMAE_scratch	n/a	n/a	mlp	stim_level	0.1095	0.0493	0.7544	0.0205	0.8918	0.0381	0.9442	0.0200	5
15	DINOv2_scratch	n/a	n/a	mlp	stim_level	0.0921	0.0372	0.8697	0.2398	1.2052	0.5933	1.0762	0.2421	5
16	DINOv2_scratch	n/a	n/a	linear	stim_level	0.0828	0.0595	1.6309	1.6660	9.5235	18.3974	2.2039	2.4151	5
17	CLIP_scratch	n/a	n/a	mlp	stim_level	0.0804	0.0649	0.7798	0.0730	0.9550	0.1676	0.9744	0.0829	5
18	CLIP_scratch	n/a	n/a	linear	stim_level	0.0526	0.0514	0.8596	0.2707	1.3049	0.9998	1.0922	0.3742	5

3 SwiFT padding ablation – detailed breakdown

The aggregate padding ablation in the main report averages across init, head, and mode. Table 6 below shows the same data broken out by every combination, in case patterns differ between subject-conditional and pooled modes or between heads.

Table 6: SwiFT NewE96 padding ablation, broken out by init, head, and mode. Each cell is the mean test_main across 5 folds (and 5 subjects for per_subject mode).

Task	Init	Head	Mode	mean	replicate	zero	spatial_only	cyclic_replicate
A_binary	resting	linear	per_subject	0.5818	0.5567	0.5873	0.5889	0.5649
A_binary	resting	linear	pooled	0.6067	0.5734	0.6055	0.6121	0.5972
A_binary	resting	mlp	per_subject	0.5405	0.5300	0.5643	0.5444	0.5344
A_binary	resting	mlp	pooled	0.5779	0.5326	0.5840	0.5797	0.5588
A_binary	scratch	linear	per_subject	0.5795	0.5550	0.5825	0.5798	0.5777
A_binary	scratch	linear	pooled	0.5791	0.5315	0.5642	0.5809	0.5771
A_binary	scratch	mlp	per_subject	0.5598	0.5358	0.5650	0.5532	0.5540

Task	Init	Head	Mode	mean	replicate	zero	spatial_only	cyclic_replicate
A_binary	scratch	mlp	pooled	0.5596	0.5166	0.5703	0.5655	0.5510
A_reg	resting	linear	per_subject	0.1166	0.0791	0.1071	0.1217	0.1073
A_reg	resting	linear	pooled	0.1538	0.1123	0.1414	0.1591	0.1447
A_reg	resting	mlp	per_subject	0.0490	0.0349	0.0917	0.0718	0.0392
A_reg	resting	mlp	pooled	0.1059	0.0535	0.1203	0.1128	0.0914
A_reg	scratch	linear	per_subject	0.0672	0.0591	0.0690	0.0656	0.0651
A_reg	scratch	linear	pooled	0.0873	0.0427	0.0900	0.0892	0.0814
A_reg	scratch	mlp	per_subject	0.0711	0.0670	0.1137	0.0855	0.0891
A_reg	scratch	mlp	pooled	0.0912	0.0482	0.0994	0.0884	0.0845
V_binary	resting	linear	per_subject	0.6711	0.6242	0.6649	0.6696	0.6677
V_binary	resting	linear	pooled	0.6860	0.6462	0.6884	0.6848	0.6835
V_binary	resting	mlp	per_subject	0.5986	0.5727	0.5999	0.6037	0.5899
V_binary	resting	mlp	pooled	0.6163	0.5824	0.5949	0.5994	0.6180
V_binary	scratch	linear	per_subject	0.6571	0.6500	0.6546	0.6544	0.6537
V_binary	scratch	linear	pooled	0.6349	0.6250	0.6314	0.6303	0.6331
V_binary	scratch	mlp	per_subject	0.6315	0.6251	0.6215	0.6135	0.6250
V_binary	scratch	mlp	pooled	0.6187	0.6095	0.6070	0.6113	0.6233
V_reg	resting	linear	per_subject	0.2640	0.1990	0.2564	0.2647	0.2618
V_reg	resting	linear	pooled	0.2561	0.2042	0.2510	0.2596	0.2559
V_reg	resting	mlp	per_subject	0.1306	0.0798	0.1387	0.1392	0.1340
V_reg	resting	mlp	pooled	0.1772	0.1221	0.1346	0.1625	0.1672
V_reg	scratch	linear	per_subject	0.1867	0.1508	0.1950	0.1830	0.1896
V_reg	scratch	linear	pooled	0.1795	0.1541	0.1835	0.1729	0.1828
V_reg	scratch	mlp	per_subject	0.1878	0.1748	0.1816	0.1926	0.1830
V_reg	scratch	mlp	pooled	0.1890	0.1646	0.1514	0.1762	0.1693